

Digital Enablement Dashboard

Project Showcase & Technical Documentation

Project Overview

A high-performance, data-driven analytics platform designed to provide real-time visibility into team delivery, project health, and executive-level KPIs. This dashboard transforms fragmented Excel-based tracking into a centralized, interactive visual hub.

The Digital Enablement Dashboard (also known as Dashboard Looker) is a sophisticated enterprise tool that bridges the gap between manual spreadsheets and automated reporting. It provides two primary layers of visibility:

- **Operational Layer:** Real-time standup tracking, bandwidth monitoring, and blocker management for delivery teams.
- **Executive Layer:** Secure, JWT-protected analytics covering financial health (SOW/PO), quality metrics (FTR), and resource capacity (Skill Matrix/KPIs).

Core Features

Operational Excellence

- **Smart Standup Panels:** Automatically extracts 'Yesterday' and 'Today' tasks, categorizing them into Deliverables vs. Operational work.
- **Visual Bandwidth Tracking:** Real-time percentage bars for team capacity, identifying bottlenecks or available resources at a glance.
- **Live Blocker Management:** Centralized tracking for QBR dates and project-critical dependencies.
- **Presentation Mode:** A specialized 'TV-optimized' UI for team standups and screen sharing.

Executive Intelligence (Manager Portal)

- **Secure Access:** Role-based access control via JWT (JSON Web Tokens).
- **Project Health Matrix:** Automated scoring system that computes 'Health' by cross-referencing multiple data sources.
- **Financial & SOW Tracking:** Interactive doughnut charts and summaries for PO status and project-wise financials.
- **KPI Scorecards:** Clickable cards for key performance indicators with deep-dive trend analysis.
- **FTE & Resource Trends:** Time-series visualization of team growth and allocation.

Connectivity & Integration

- **Multi-Source Support:** Seamlessly connects to Local Excel files, Google Sheets (live export), and SharePoint/OneDrive links.

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- **Zero-Delay Refresh:** Built-in polling mechanism keeps data fresh without requiring manual page reloads.

Technical Stack

Frontend

- **Vanilla JS (ES6+):** High-performance, zero-framework architecture for lightning-fast load times.
- **Modern CSS3:** Custom grid/flexbox layouts with CSS variables for dynamic theming.
- **Chart.js:** Interactive, high-fidelity data visualizations.

Backend

- **Node.js & Express:** Scalable REST API architecture.
- **JWT Authentication:** Secure session management for executive features.
- **Multer:** Robust file handling for Excel uploads.

Data Engineering (The 'Secret Sauce')

- **Python Normalizer:** A specialized bridge using Pandas and OpenPyXL to handle complex Excel logic.
- **Smart Header Detection:** Scans up to 30 rows deep to find headers, overcoming 'messy' human-edited spreadsheets.
- **Fuzzy Matching:** Uses string similarity algorithms to map column names (e.g., matching 'PM' to 'Project Manager') automatically.
- **Hybrid Parsing:** Seamless fallback to Node.js parsers if Python environments are unavailable.

Architecture

The system architecture is divided into three distinct layers ensuring data integrity and user security:

- **Data Sources:** Local Excel, Google Sheets, SharePoint / OneDrive
- **Backend (Node.js):** Express Server, JWT Auth Middleware, execDataService
- **Parsing Engine:** Python Normalizer (Smart Parse) with Node.js Fallback Parser
- **Frontend (Vanilla JS):** Operational Dashboard, Executive Portal, Data Health Diagnostics

Key Technical Highlights (Showcase)

Intelligent Parsing Engine

Traditional dashboards break when a user adds a row or renames a column in the source Excel. This project

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implements a Python-powered normalizer that uses fuzzy logic and deep-scanning to ensure 100% uptime regardless of spreadsheet formatting changes.

Lightweight Performance

By avoiding heavy frameworks like React or Angular, the dashboard maintains a near-zero footprint, making it ideal for low-bandwidth environments and instant-loading presentation monitors.

Enterprise Security

Features a complete authentication flow with encrypted password hashing (Bcrypt) and secure token-based sessions (JWT), ensuring sensitive financial and resource data is protected.

Data Health Diagnostics

Includes a dedicated `/api/exec/data-health` endpoint that provides a 'transparency report' on data quality, identifying missing columns or parsing errors before they impact decision-making.